ADAPTIVE BRAKE CONTROL

Adaptive Brake Control continuously monitors sensors and applies precise braking force to prevent slips, maintain stability, and avoid collisions. It checks wheel speeds during cornering, applies emergency braking when obstacles are detected, and distributes force optimally across axles based on load and road conditions.

Three-Layer Processing Architecture

The system follows the three-layer flow ,implemented through AUTOSAR components.

1. Data Collection Layer – Raw Sensor Inputs

The Data Collection Layer is responsible for continuously capturing real-time data from the vehicle and its surrounding environment. This layer acts as the primary source of information required for adaptive brake control decisions.

- **Radar sensors** measure the distance and relative speed of obstacles ahead, such as vehicles or pedestrians. They operate reliably in various weather conditions and help estimate time-to-collision.

- **Camera inputs** provide visual and environmental context by detecting and classifying objects, lane markings, traffic signs, and road edges. Cameras enhance situational awareness by distinguishing between stationary and moving objects.

- **Wheel speed sensors** monitor the rotational speed of each wheel, enabling detection of wheel slip, sudden deceleration, and traction loss. This information is critical for coordinating braking with ABS and stability control systems.

- **Vehicle speed signals**, obtained from ECU calculations, represent the overall motion of the vehicle, while **road condition inputs** (dry, wet, icy, or uneven surfaces) influence braking response and stopping distance.

2. Data Processing Layer cleans and fuses data via AUTOSAR SWCs:

The Data Processing Layer plays a critical role in converting raw vehicle and environment data into meaningful, reliable information. This layer is typically implemented using AUTOSAR Software Components (SWCs) and operates in real time to support advanced driving functions.

- Adaptive sensor filtering removes noise from raw signals
- Data fusion validates radar-camera-wheel speed correlations
- Situational awareness assesses driving context (highway,village curves roads)
- The vehicle dynamics model predicts stability limits.

i. Sensor Data Cleaning and Pre-Processing:

Raw data from sensors such as radar, camera, wheel speed sensors, IMU, and steering angle sensors often contains noise, delays, and outliers.
Adaptive sensor filtering algorithms (e.g., Kalman filters, low-pass filters, moving averages) dynamically adjust filtering parameters based on vehicle speed and road conditions.
This improves signal stability and accuracy, ensuring robust inputs for higher-level control logic.

ii. Multi-Sensor Data Fusion:

Data fusion SWCs combine inputs from radar, camera, wheel speed, and yaw rate sensors to create a unified and consistent representation of the vehicle's surroundings.
Cross-validation techniques verify correlations (e.g., radar object distance vs. camera object classification).

Redundant sensor checks help detect faulty sensors and increase functional safety (ISO 26262 compliance).

Time synchronization ensures all sensor data is aligned using a common time base.

iii. Situational Awareness and Context Detection:

Situational awareness SWCs analyze fused data to understand the driving environment:

- Highway driving
- Urban/village roads
- Curved or hilly roads
- Stop-and-go traffic conditions

The system identifies lane geometry, road curvature, traffic density, and weather effects. This context information allows the vehicle to adapt its behavior (e.g., adjust following distance or steering sensitivity).

iv. Vehicle Dynamics Modeling:

The vehicle dynamics model estimates longitudinal, lateral, and vertical vehicle behavior.

It predicts:
1. Tire slip and grip limits
2. Vehicle stability boundaries
3. Roll, yaw, and pitch behavior

These predictions help determine safe operating limits for braking, acceleration, and steering.

v. Output to Decision and Control Layers:

Processed and validated data is provided to:

1. ADAS decision-making SWCs
2. Vehicle control SWCs (ESC, ACC, AEB, LKA)

Communication occurs through AUTOSAR RTE, ensuring standardized and deterministic data exchange.
Diagnostic and status signals are also forwarded for monitoring and fault handling.

vi. Benefits of the Data Processing Layer:

Improved accuracy and reliability of sensor data

Enhanced safety and fault tolerance

Faster and more stable control decisions

Scalability for future autonomous driving functions

3. Predictive Analytics Layer:

(AI-Based Decision Making in Adaptive AUTOSAR)

The **Predictive Analytics Layer** is the **brain** of the Adaptive Brake Control System. It uses the processed data from **Layer 2** to **predict danger** and decide **how the brakes should work**. This layer runs as high-speed applications in **Adaptive AUTOSAR**.

i. Risk Assessment Model

This part **continuously checks the driving situation**.

It uses information like:

- Distance to the vehicle or object in front
- Relative speed (how fast we are getting closer)
- Road condition (dry, wet, slippery)
- Driver actions (accelerating or braking)

Using simple AI and math models, it calculates:

> **Collision Probability** – how likely an accident will happen if nothing is done.

ii. Braking Strategy Calculation

If danger is detected, the system decides:

- **How strong** the braking should be
- **How fast** the brakes should be applied

It calculates:

- Required slowing down (deceleration)
- Brake pressure
- Brake timing

The braking force is adjusted based on:

- Vehicle speed
- Distance to the obstacle
- Road grip

This makes sure the braking is **smooth when possible and strong when necessary**.

iii. Adaptive Brake Decision

Based on the risk level, the system chooses **one of these actions**:

Situation	Decision
Low risk	No braking
Medium risk	Soft braking
High risk	Emergency braking

This decision is sent **immediately** to the brake control system.

iv. Brake Actuator Command

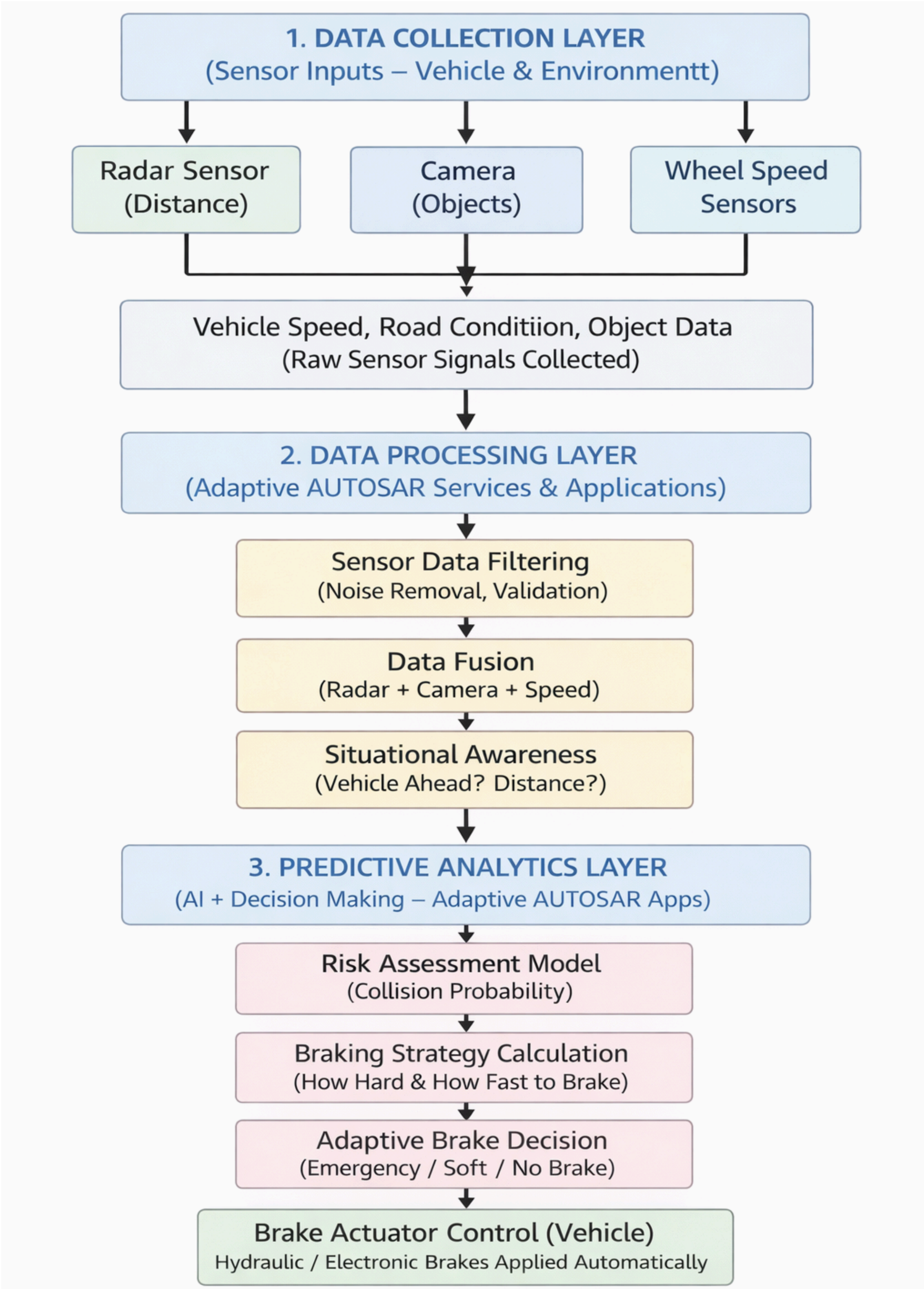
Finally, the command is sent to:

- **Electronic Brake Control Unit (EBCU)**
- **Brake actuators (hydraulic or electronic)**

Then the car:

- Applies brakes automatically
- Reduces speed safely
- Avoids or reduces the accident

FLOWCHART :


1. DATA COLLECTION LAYER
(Sensor Inputs – Vehicle & Environmentt)
Radar Sensor
(Distance)
Camera
(Objects)
Wheel Speed
Sensors
Vehicle Speed, Road Conditiion, Object Data
(Raw Sensor Signals Collected)
2. DATA PROCESSING LAYER
(Adaptive AUTOSAR Services & Applications)
Sensor Data Filtering
(Noise Removal, Validation)
Data Fusion
(Radar + Camera + Speed)
Situational Awareness
(Vehicle Ahead? Distance?)
3. PREDICTIVE ANALYTICS LAYER
(AI + Decision Making – Adaptive AUTOSAR Apps)
Risk Assessment Model
(Collision Probability)
Braking Strategy Calculation
(How Hard & How Fast to Brake)
Adaptive Brake Decision
(Emergency / Soft / No Brake)
Brake Actuator Control (Vehicle)
Hydraulic / Electronic Brakes Applied Automatically